DIPLOMA PROGRAMME PRACTICE

Mathematics

Applications and interpretation SL

Set B / Paper 1 / Short response

90 minutes · 80 marks · All questions compulsory

Use a graphic display calculator and a clean Mathematics formula booklet.

Write your reasoning and working in the spaces provided; use additional answer paper if needed. Unless stated otherwise, give exact answers or round numerical results to three significant figures.

Diagrams are illustrative unless coordinates or dimensions are specified.

QUESTION 1

6 marks

A device stores 7.2×10^9 bytes of observations in a day. Each full archive block stores exactly 3.0×10^6 bytes. For this calculation take the quoted values as exact.

(a) Find the number of full blocks required for one day. Give your answer in ordinary notation. [2]

(b) Compression is expected to save 18% of the bytes, but the actual saving could be from 16% to 20%, inclusive. Find the smallest guaranteed sufficient number of blocks and explain your choice of percentage. [4]

QUESTION 2

6 marks

A battery delivers 84 km of range when new. At each annual inspection its measured range is modelled as 93% of the previous inspection's range.

(a) Write the model after n annual intervals and find the predicted range after three. [2]

(b) Replacement is required at the first inspection when range is below 55 km. Determine which inspection triggers replacement, justifying the integer choice. [4]

QUESTION 3

6 marks

From a lookout, an observer sights a buoy on a bearing of 040° and a second buoy on a bearing of 105°. The buoys are 2.8 km and 4.1 km from the lookout respectively.

(a) Find the included angle and the distance between the buoys. [3]

(b) A boat travels directly from the first buoy to the second at 12 km/h. Find the time in minutes. Explain why the result might underestimate the actual time. [3]

QUESTION 4

6 marks

Forty calls are grouped by duration t minutes. The frequencies for $0 \le t < 2$, $2 \le t < 4$, $4 \le t < 8$ and $8 \le t < 12$ are 6, 14, 12 and 8.

(a) Estimate the mean call duration. [3]

(b) What fraction of calls lasted at least 4 minutes? Explain why this is exact for the sample although your mean is an estimate. [3]

QUESTION 5

6 marks

The height of a launched water jet is $h(t) = 0.8 + 6t - 4.9t^2$ metres, where t is time in seconds after launch. The model ends when the jet first reaches the ground.

(a) Find the maximum height and the time at which it occurs. [3]

(b) Determine the model's time domain, and explain why the other algebraic zero is excluded. [3]

QUESTION 6

6 marks

A factory models fill mass X by a normal distribution of mean 506 g and standard deviation 3.8 g. Underweight packs have mass below 500 g.

(a) Find the probability that a randomly selected pack is underweight. [2]

(b) Find the 95th percentile of mass. A supervisor claims exactly 5 packs in every batch of 100 will exceed that mass. Comment. [4]

QUESTION 7

7 marks

Five film clips P, Q, R, S and T receive two sets of ranks (1 = best). The youth panel ranks them 1, 2, 3, 4, 5; the adult panel ranks them 2, 4, 1, 3, 5, in the same clip order.

(a) Calculate Spearman's rank correlation coefficient. [4]

(b) Explain why ranking observations can hide information. Does the coefficient show that both panels chose the same winner? [3]

QUESTION 8

7 marks

A cylindrical sample tube has internal radius 1.6 cm and a straight usable length of 11 cm. It may be filled to only 85% of its cylindrical volume. Treat the stated dimensions and fill fraction as exact in this model. Use 1 cm^3 = 1 mL.

(a) Determine its permitted liquid volume. [3]

(b) A technician must store 1.88 litres. Find the minimum number of tubes, and explain why rounding the capacity first to the nearest 10 mL could give a different decision. [4]

QUESTION 9

7 marks

Two mobile clinics are at U(−2, 1) and V(4, 1). All coordinates are in kilometres; travel is modelled along straight lines.

(a) Find the boundary between their nearest-clinic regions. Explain why it is vertical. [3]

(b) A river makes the actual journey from P(0, 3) to U twice the straight-line distance, while the journey to V equals its straight-line distance. Determine which clinic is now closer by journey distance. [4]

QUESTION 10

7 marks

At a fair a player pays 4 dollars and receives exactly one gross prize: 0 dollars with probability 0.60, 5 dollars with probability 0.30 or 20 dollars otherwise.

(a) Find the probability of the largest prize and the expected net gain to the player. [3]

(b) The fair keeps the fee and probabilities fixed. Find the largest prize that would make the game fair, and explain what fair means here. [4]

QUESTION 11

8 marks

The energy E in a small thermal store after t hours is modelled by $E(t) = 18 + 8t - t^2$ kWh, for $0 \le t \le 5$.

(a) Find E(2) and the slope of the tangent to the graph at t = 2. [3]

(b) Use the tangent to estimate E(2.5), and calculate the error in your estimate. [3]

(c) Explain geometrically why the same tangent also overestimates E(1.5). [2]

QUESTION 12

8 marks

Independent random samples of delivery times are taken for two firms. Assume approximately normal populations. A two-sided independent-samples t-test of equal population means returns p = 0.031.

(a) State both hypotheses in the context of this study. [2]

(b) State the decision and its contextual conclusion at 5%, then explain what changes at 1%. [4]

(c) Explain why the reported p-value alone cannot establish which firm is faster or whether the difference matters to customers. [2]

MARKING AND REVIEW

Worked solutions

Applications and interpretation SL

Set B / Paper 1 / Short response

Maximum mark: 80. The allocations are for independent self-assessment.

M = method, A = accuracy or stated result, R = reasoning. Award equivalent valid methods. Follow through from an earlier numerical slip when the subsequent method is correct and the task is not simplified.

Keep unrounded calculator values until the final answer. Accept sensible equivalent accuracy unless the question specifies otherwise.

Read the student's written interpretation as well as their numerical result. A calculator output alone may not justify a conclusion.

SOLUTION 1

6 marks

(a) Find the number of full blocks required for one day. Give your answer in ordinary notation. [2]

M1 Divide 7.2×10^9 by 3.0×10^6.

A1 2400 blocks.

(b) Compression is expected to save 18% of the bytes, but the actual saving could be from 16% to 20%, inclusive. Find the smallest guaranteed sufficient number of blocks and explain your choice of percentage. [4]

R1 The smallest saving gives the largest file, so use 16%.

M1 Multiply 2400 by 0.84.

A1 2016 blocks.

R1 This already is a whole number; any smaller allocation fails at the 16% endpoint.

SOLUTION 2

6 marks

(a) Write the model after n annual intervals and find the predicted range after three. [2]

A1 $R(n) = 84 \times 0.93^n$.

A1 R(3) = 67.6 km (3 sf).

(b) Replacement is required at the first inspection when range is below 55 km. Determine which inspection triggers replacement, justifying the integer choice. [4]

M1 Compare the model with 55, using a table or solving for n.

A1 R(5) = 58.4 km (3 sf).

A1 R(6) = 54.3 km (3 sf).

R1 It first falls below 55 at inspection 6 because the sequence decreases.

SOLUTION 3

6 marks

(a) Find the included angle and the distance between the buoys. [3]

A1 Included angle = 105° − 40° = 65°.

M1 $d^2 = 2.8^2 + 4.1^2 - 2(2.8)(4.1)\cos 65°$.

A1 d = 3.87 km.

(b) A boat travels directly from the first buoy to the second at 12 km/h. Find the time in minutes. Explain why the result might underestimate the actual time. [3]

M1 Time = 60d/12.

A1 19.3 minutes.

R1 A detour around a shoal or an opposing current could increase time; accept an appropriate contextual reason.

SOLUTION 4

6 marks

(a) Estimate the mean call duration. [3]

M1 Use class midpoints 1, 3, 6, 10.

M1 Weight by frequency: (6 + 42 + 72 + 80)/40.

A1 5.00 minutes.

(b) What fraction of calls lasted at least 4 minutes? Explain why this is exact for the sample although your mean is an estimate. [3]

A1 (12 + 8)/40 = 1/2.

R1 The threshold lies on a class boundary, so every call can be counted on the correct side.

R1 The mean used midpoints in place of the unknown individual durations.

SOLUTION 5

6 marks

(a) Find the maximum height and the time at which it occurs. [3]

M1 Use the maximum of the quadratic on $t \geq 0$.

A1 $t = 0.612$ s.

A1 Height = 2.64 m.

(b) Determine the model's time domain, and explain why the other algebraic zero is excluded. [3]

M1 Solve $h(t) = 0$.

A1 $0 \leq t \leq 1.35$ s (approximately).

R1 The other root is before launch and outside the physical experiment.

SOLUTION 6

6 marks

(a) Find the probability that a randomly selected pack is underweight. [2]

M1 Use the lower normal tail with mean 506 and standard deviation 3.8.

A1 0.0572.

(b) Find the 95th percentile of mass. A supervisor claims exactly 5 packs in every batch of 100 will exceed that mass. Comment. [4]

M1 Use inverse normal with lower area 0.95.

A1 512 g.

R1 A randomly selected pack has probability 0.05 of exceeding this threshold.

R1 Five is an expected number, not an exact batch count.

SOLUTION 7

7 marks

(a) Calculate Spearman's rank correlation coefficient. [4]

M1 Form paired rank differences −1, −2, 2, 1, 0.

A1 Sum of squared differences = 10.

M1 Use $1 - 6 \times 10/[5(5^2 - 1)]$, or correlation of the two rank lists.

A1 Coefficient = 0.5.

(b) Explain why ranking observations can hide information. Does the coefficient show that both panels chose the same winner? [3]

R1 Rankings discard the gaps in actual scores.

R1 No; a positive coefficient describes overall association, not identical first choices.

A1 The youth panel selects P, while the adult panel selects R.

SOLUTION 8

7 marks

(a) Determine its permitted liquid volume. [3]

M1 $V = \pi(1.6)^2(11)$.

A1 Full volume = 88.5 mL (3 sf).

A1 Permitted volume = 75.2 mL (3 sf).

(b) A technician must store 1.88 litres. Find the minimum number of tubes, and explain why rounding the capacity first to the nearest 10 mL could give a different decision. [4]

M1 Convert to 1880 mL and divide by the unrounded permitted capacity.

A1 The quotient is approximately 25.001; 25 tubes hold only about 1879.93 mL.

A1 At least 26 tubes are required.

R1 Rounding each capacity to 80 mL overstates the total and could suggest only 24 tubes.

SOLUTION 9

7 marks

(a) Find the boundary between their nearest-clinic regions. Explain why it is vertical. [3]

A1 Midpoint = (1, 1).

R1 UV is horizontal; its perpendicular bisector is vertical.

A1 Boundary x = 1.

(b) A river makes the actual journey from P(0, 3) to U twice the straight-line distance, while the journey to V equals its straight-line distance. Determine which clinic is now closer by journey distance. [4]

M1 Straight distances to U and V are $\sqrt{8}$ and $\sqrt{20}$.

A1 Adjusted distances are $2\sqrt{8} \approx 5.66$ and $\sqrt{20} \approx 4.47$ km.

A1 V is now closer.

R1 The straight-line boundary would have assigned P to U because $x < 1$; the river breaks the model assumption.

SOLUTION 10

7 marks

(a) Find the probability of the largest prize and the expected net gain to the player. [3]

A1 Remaining probability = 0.10.

M1 Expected gross prize = 0(0.60) + 5(0.30) + 20(0.10).

A1 Expected net gain = 3.50 − 4 = −0.50 dollars.

(b) The fair keeps the fee and probabilities fixed. Find the largest prize that would make the game fair, and explain what fair means here. [4]

M1 Let the new top prize be k. Set expected net gain to zero.

A1 1.50 + 0.10k = 4.

A1 k = 25 dollars.

R1 Fair means zero expected net gain over repeated plays, not that every player breaks even.

SOLUTION 11

8 marks

(a) Find E(2) and the slope of the tangent to the graph at t = 2. [3]

A1 E(2) = 30 kWh.

M1 E′(t) = 8 − 2t.

A1 E′(2) = 4 kWh per hour.

(b) Use the tangent to estimate E(2.5), and calculate the error in your estimate. [3]

M1 Tangent equation E = 30 + 4(t − 2).

A1 Estimate = 32 kWh.

A1 Actual E(2.5) = 31.75 kWh; overestimate = 0.25 kWh.

(c) Explain geometrically why the same tangent also overestimates E(1.5). [2]

R1 The graph is concave down on the interval.

R1 It lies below its tangent on both sides of t = 2, apart from the contact point.

SOLUTION 12

8 marks

(a) State both hypotheses in the context of this study. [2]

A1 H0: the firms have equal population mean delivery times.

A1 H1: their population mean delivery times differ.

(b) State the decision and its contextual conclusion at 5%, then explain what changes at 1%. [4]

M1 Compare 0.031 with 0.05.

A1 Reject H0 at 5%; there is evidence of a difference in the firms' population means.

M1 Compare 0.031 with 0.01.

A1 At 1% do not reject H0; the evidence is insufficient at that level.

(c) Explain why the reported p-value alone cannot establish which firm is faster or whether the difference matters to customers. [2]

R1 A two-sided p-value has no direction; compare the observed means to establish the sample direction.

R1 It does not give the size of the time difference; assess a difference in minutes against a meaningful threshold.